

1 EX-002 — screw/barrel/die

물리 검증·안전 인증·통전 승인이 아니다. E-stop, lid/service interlock, branch fuse, 독립 thermal fuse를 정상 firmware와 독립 구현하고 exact received component 정격·배선·보호소자를 실측 확인하기 전 통전하지 않는다.

전체 배치: GGM-FULL-ASM / 구동계: GGM R2. 기본 조립 투상도는 통합 전 참조이며 최신 전체 배치가 아니다. Revision: final-design-fabrication-closure-v0.8 · 상태: DIGITAL_DOCUMENT / PHYSICAL_NOT_RUN / USER_APPROVAL_REQUIRED

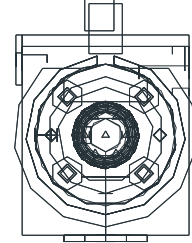
EX-002 screw barrel die

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 15

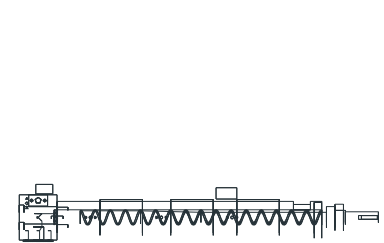
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 44.00 · Y 380.50 · Z 60.50 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

1.1 제작·검사 기준

- 단위: mm · 제3각법 · 축척: NTS(기입 치수 우선)
- 재료/구성: nitrided SCM440 / NSK 51102 / 17-4PH die insert
- 핵심 치수/공차: cold screw/barrel diametral clearance 0.28–0.32 mm; 51102 pocket clearance 0.30–0.35 mm; shaft abutment ≥ 23 mm; housing abutment ≤ 20 mm; shim-set loaded-direction endplay 0.05–0.15 mm; coaxiality ≤ 0.05 mm; thrust M6 $\times$ 8 = 9 N·m; die SYS-04 M4 $\times$ 45 $\times$ 4 stock screws cut/deburred to 42.5 ± 0.1 give engagement 6.82–7.40/thread-bottom clearance 0.60–1.18; digital torque 1.50 N·m, physical receipt/leak/first thermal cycle NOT_RUN; retainer M4 $\times$ 2 = 1.2 N·m
- 일반공차: ISO 2768-m. 개별 부품은 exports/final/manufacturing 도면과 interface_catalog.csv를 함께 검사한다.
- Source: 93ec6a9c0c2afa3ad72288a967f07b7c8d03c99e · 물리 검증: NOT_RUN